

Hypobaric physiology in air transport of dogs and cats: pressurization, oxygen and physiological response in commercial flight

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Abstract

Objective. To analyze the physiological compensation mechanisms and reserve limits in the face of moderate hypoxia in canids and felids during commercial air transport. **Method.** Technical review of comparative respiratory physiology, gas physics principles (Boyle's Law, Dalton's Law), FAA/EASA/LAR regulations on cabin pressurization, and available evidence on in-flight physiological response. **Results.** Commercial cabin pressurization equivalent to 6,000-8,000 feet reduces PiO_2 to 15.1-17.1 kPa. Canids and felids compensate through hyperventilation, cardiac output redistribution, and HPA axis activation; reserve limits narrow critically in brachycephalic breeds, geriatric animals, and heat stress conditions. Prolonged exposure without water access aggravates hypocapnia-induced dehydration. **Conclusion.** Air transport generates moderate, documentable hypoxia and physiological stress; pre-flight assessment and travel condition selection are determinants of animal welfare.

Keywords: moderate hypoxia, pet air transport, brachycephalic, cabin pressurization, PiO_2 , comparative respiratory physiology, FAA, EASA, animal welfare in flight

Section 1 The hypobaric environment in commercial aviation

1.1 What pressurization is and why it exists

As an aircraft climbs in altitude, exterior atmospheric pressure decreases according to the relationship defined by the International Standard Atmosphere (ISA). At 10,000–12,000 m cruise altitude —typical range for long-haul commercial flights— exterior barometric pressure is so low that available oxygen concentration would be incompatible with survival without protection equipment. To address this, commercial aircraft are fitted with pressurization systems that compress exterior air and introduce into the cabin a mixture at higher pressure than outside, maintaining physiologically tolerable conditions.

Pressurization does not reproduce sea-level conditions exactly. For structural design reasons —maintaining a high pressure differential between interior and exterior requires significant structural strength and greater weight— aircraft are pressurized to an equivalent altitude that lies, under normal operating conditions, between 1,500 and 2,438 m above sea level.

Regulatory basis: The FAA (14 CFR Part 25.841) establishes that cabin pressure altitude must not exceed 8,000 ft (2,438 m) under normal operating conditions. EASA (CS-25) harmonizes with this same standard. These are design limits; in actual operation, cabin altitude may be lower, depending on the aircraft and cruise altitude.

1.2 Passenger cabin vs. climate-controlled cargo hold

An operationally relevant distinction exists between the passenger space and cargo holds. The lower holds of modern commercial aircraft are divided into compartments; those designated for live animal transport («Live Animal Compartment» in IATA nomenclature) are pressurized and climate-controlled compartments that generally share the same pressurization system as the passenger cabin.

However, relevant differences exist in the physical environment of the cargo hold:

- Temperature control: although temperature is regulated, the hold may experience wider variations than the cabin, especially during boarding and disembarking phases on the tarmac.
- Relative humidity: air at high altitude has very low humidity. After compression for pressurization, relative humidity in cabin and hold is typically very low (10–20%).
- Noise level: cargo holds are exposed to higher noise levels than the passenger cabin. This is an additional stressor for animals.
- Absence of light cycle: as analysed in Article 7 of this series, the hold lacks a natural light/dark cycle.

The distinction between a pressurized/climate-controlled hold compliant with IATA LAR standards and an unconfigured hold is essential: IATA welfare requirements (51st ed., 2025) demand minimum ventilation, temperature and space conditions that are not guaranteed on all aircraft or all flights.

Section 2 Physical basis: pressure, oxygen and altitude

2.1 Barometric pressure: sea level vs. 2,438 metres

Barometric pressure at sea level is, by definition in the ISA, 101,325 kPa (760 mmHg). At 2,438 m (8,000 ft), barometric pressure is approximately 74.7 kPa (560 mmHg), representing a reduction of approximately 26% relative to sea level. This reduction does not alter the percentage composition of air—which remains approximately 21% oxygen at any altitude—but it does reduce the partial pressure of each component gas in direct proportion to the total pressure drop.

2.2 Partial pressure of oxygen: the physiologically relevant parameter

Dalton's law states that in a gas mixture, the partial pressure of each component is proportional to its molar fraction and the total pressure of the mixture. Inspired partial pressure of oxygen (P_{iO_2}) is calculated as: $P_{iO_2} = F_{iO_2} \times (P_b - P_{H_2O})$, where F_{iO_2} is the oxygen fraction in inspired air (≈ 0.21), P_b is barometric pressure and P_{H_2O} is partial pressure of water vapour at body temperature (approximately 6.3 kPa / 47 mmHg in mammals). At equivalent cabin altitude of 2,438 m, P_{iO_2} is significantly lower than at sea level. West (2017) describes how this reduction affects the «oxygen cascade»: the series of steps from inspired air to the mitochondrion.

2.3 The oxyhaemoglobin dissociation curve and the inflection point

Peacock (1998) notes that the oxyhaemoglobin dissociation curve has a sigmoidal shape, with a relatively flat upper segment and a steep lower segment. The inflection zone lies approximately between a P_{aO_2} of 60–80 mmHg. At sea level, a healthy mammal operates on the flat upper segment. However, when P_{aO_2} falls toward or below the inflection zone, small additional reductions produce disproportionately large drops in saturation. Equivalent cabin altitude of 2,438 m places healthy mammals in a zone where ventilatory compensation is sufficient. Animals with conditions that shift the curve to the right (acidosis, hyperthermia, hypercapnia from airway obstruction) may find themselves in a more unfavourable position.

Section 3 Physiological response in healthy animals

3.1 Ventilatory compensation

The primary response to a reduction in P_{iO_2} is an increase in minute ventilation, mediated by peripheral chemoreceptors (carotid bodies). This mechanism, termed the hypoxic ventilatory response (HVR), increases respiratory rate and/or depth. Davis and Cummings (2019) review compensatory ventilatory response in dogs, confirming that the HVR mechanism is present and functionally equivalent to that described in other mammals. The counterpart is relative hypocapnia: mild respiratory alkalosis is an expected and self-limiting finding in healthy animals at moderate altitudes.

3.2 Mild cardiovascular changes

Increased cardiac output: through elevation of heart rate and/or stroke volume. Flow redistribution: vasoconstriction in non-priority territories and vasodilation in high-demand territories (brain, myocardium). Borgstrom et al. (1975) documented compensatory cerebral vasodilation in response to moderate hypoxia in dogs. Long-term haematological response: Smith et al. (1987) documented increased erythropoietin in dogs with prolonged exposure; however, a commercial flight of hours is not sufficient to trigger a significant haematological response.

3.3 Physiological reserve as the central concept

The most important concept is not the magnitude of the hypoxic stimulus itself—which is moderate and well tolerated by most healthy mammals— but that of physiological reserve : the organism's capacity to respond to additional demand beyond its baseline conditions. A healthy animal at rest at sea level operates well below its maximum capacity. When exposed to equivalent cabin altitude, the reduction in PiO_2 activates part of that reserve. If the animal already has part of its reserve compromised by a pre-existing condition—cardiac disease, airway obstruction, anaemia—the margin available for hypobaric compensation is smaller.

Section 4 Situations where compensatory capacity may be limited

4.1 Brachycephalic Obstructive Airway Syndrome (BOAS)

Brachycephalic breeds—English Bulldog, French Bulldog, Pug, Boston Terrier, Shih Tzu, Boxer, and their feline analogues such as the Persian— present anatomical alterations of the upper airway that generate resistance to airflow. The primary compensatory response to reduced PiO_2 is increased minute ventilation; in an animal with BOAS, the airway has increased resistance: the respiratory work required is greater, implying greater metabolic demand and greater heat production. Panting—the main heat-dissipation mechanism in dogs— also passes through the same obstructed airway. Ladlow et al. (2018) and Packer et al. (2015) documented that airway resistance in these breeds significantly limits ventilatory compensatory capacity under stress. Hall et al. (2020) documented that brachycephalic breeds have a significantly lower heat tolerance threshold.

4.2 Cardiac disease

An animal with cardiac disease limiting cardiac functional reserve may have the compensation pathway compromised. Transport stress—which activates the SAM and HPA axes— and the hypobaric environment may represent a greater load than each factor alone. Evidence note: there is no specific study that has measured the impact of commercial flight on dogs with documented cardiac disease.

4.3 Significant anaemia

Significant anaemia reduces oxygen transport capacity. Under hypobaric conditions, the animal has both reduced transport capacity (from anaemia) and reduced haemoglobin loading pressure (from lower PiO_2). Niskanen et al. (2016) provide reference values for arterial gases in dogs at sea level. The inference on anaemia and hypobaria is based on principles of oxygen transport physiology.

4.4 Extreme age: puppies and geriatric animals

Very young puppies: ventilatory regulation and peripheral chemoreceptor response are not fully mature. Geriatric animals: cardiac and respiratory reserve decrease with age. Subclinical pathologies common in advanced age may reduce the margin for compensation. No specific studies exist on commercial flight in puppies or geriatric animals.

4.5 Elevated body condition (obesity)

Mechanical restriction of ventilation: excess adipose tissue increases respiratory work by reducing thoracic compliance. Increased metabolic demand: greater body mass implies greater absolute oxygen consumption. Obesity in dogs frequently coexists with brachycephalic breeds and cardiac disease.

Section 5 Hypobaria and thermoregulation

5.1 Panting as the main canine heat-loss mechanism

The dog depends primarily on panting for heat dissipation. At lower barometric pressure, air density is lower; the convection coefficient decreases with density. Heat loss efficiency by convection and evaporation may be slightly reduced in the hypobaric cabin environment relative to sea level—a real physical effect but of modest magnitude.

5.2 Relative humidity and mucosal dehydration

Relative humidity in cabin and hold is typically 10–20%. Consequences: increased insensible water loss—panting in very dry air implies greater respiratory water loss—and respiratory mucosal desiccation.

5.3 Tarmac thermal stress: the best-documented risk

The literature consistently identifies the period of stay on the tarmac—boarding, ground waits, disembarking—as that of highest thermal stress risk. Hall et al. (2020) documented that confinement and elevated temperature are the main trigger of heat stroke in dogs. Vanthana et al. (2023) reviewed the impact of confinement and humidity on canine panting efficiency. Risk of severe thermal stress on the tarmac is greater than the risk derived from hypobaria during flight itself for healthy animals.

Section 6 Boyle's law and gas expansion at altitude

6.1 The physical principle

Boyle's law states that, at constant temperature, the volume of a fixed mass of gas is inversely proportional to its pressure: $PV = \text{constant}$. At 2,438 metres, pressure is approximately 26% lower than at sea level; a mass of gas will occupy approximately 35% more volume. Ernsting's Aviation Medicine (2023) describes this principle as fundamental to barophysiology in human aviation medicine.

6.2 Gastrointestinal gas expansion

The gastrointestinal tract contains gas as a normal physiological state. During ascent to cruise altitude, this gas expands according to Boyle's law. Gowing (2015) mentions the need to avoid diets that favour intestinal gas production in the hours before flight. In most healthy animals, expansion produces mild distension that resolves with increased peristalsis.

6.3 Situations of greater clinical relevance

Recent abdominal or thoracic surgery: may leave residual gas that, when expanding at altitude, may compromise adjacent structures. Gastric dilatation: breeds predisposed to gastric dilatation-volvulus (GDV)—Great Dane, Doberman, German Shepherd—have potentially greater baseline gastric gas accumulation. Additional expansion at altitude is biologically coherent and justifies caution in dietary management prior to transport.

6.4 Difference from hyperbaric phenomena

Hyperbaric phenomena, associated with scuba diving or hyperbaric chambers, involve pressures far above sea level and pathophysiological mechanisms (decompression sickness, gas embolism) that are not applicable to the commercial flight context.

Section 7 Current state of the evidence

7.1 Honest declaration: what science knows and what it does not

7.2 The five most relevant evidence gaps

(1) In situ physiological monitoring: no study has measured oxygen saturation, arterial PCO₂ or cardiorespiratory parameters in dogs or cats inside a cargo hold during an actual commercial flight. (2) Quantification of intestinal gas expansion in dogs: no imaging studies have measured abdominal distension volume. (3) Specific feline physiological response to moderate hypobaria: the literature is virtually non-existent. (4) Effect of vibration on respiratory mechanics: environmental factor not studied in veterinary

context. (5) Acclimatization kinetics in short- and medium-duration flights: no definitive answer in the veterinary literature.

Section 8 Common myths: debunked with physics

Myth 1: «The cargo hold has no oxygen» — False. Holds designated for live animals are pressurized compartments sharing the same pressurization system as the cabin. FAA and EASA regulate pressurization; IATA LAR establishes ventilation requirements.

Myth 2: «Animals run out of air on long flights» — Unfounded for certified holds. The system continuously renews air. O₂ concentration is maintained at 21%. What occurs is reduction in the partial pressure of that oxygen, which is distinct from absence of oxygen.

Myth 3: «It's the same as unpressurised military aviation» — False. A fighter at 10,000 m without pressurization exposes to lethal barometric pressures; a pressurized commercial hold operates at equivalent altitudes of 1,500 to 2,438 metres.

Myth 4: «Brachycephalic dogs always die in flight» — Incorrect and alarmist. Data do not support claims of elevated or predictable mortality. What is documented is that BOAS limits ventilatory compensatory reserve; magnitude in the individual must be assessed by a veterinarian.

Myth 5: «The hold is equivalent to a vacuum chamber» — False. A vacuum chamber has pressure close to zero. A commercial hold has pressure equivalent to 1,500–2,438 metres, normal air composition (21% O₂), regulated temperature and continuous air renewal.

Section 9 Mechanistic synthesis: integration of factors

9.1 The cumulative physiological load model

Each factor —hypobaria, stress, confinement, fasting, low humidity, noise, absence of light cycle, and on long flights circadian rhythm desynchronization (jet lag) — consumes part of the animal's physiological reserve. For a healthy adult animal with ample reserves, the sum in a 12-hour flight is manageable. For an animal whose reserve is already partially compromised, the same sum may exceed compensatory capacity.

9.2 The role of individual condition

The same flight may represent radically different physiological loads for two animals of the same species. Pre-travel veterinary evaluation is not bureaucratic procedure; it is the mechanism by which the general information in this article is translated into an informed decision for a specific animal.

9.3 Species differences in the hypobaric context

Dogs: hypoxic ventilatory response is documented and functionally effective. The main additional risk variable is BOAS prevalence. Cats: specific evidence is significantly more limited than in dogs. Extrapolation from canine or human data must be made with the methodological qualification it deserves.

Section 10 Limitations of the analysis

(1) Absence of in situ studies: no physiological monitoring studies exist during flight in the cargo hold. (2) Heterogeneity of transport conditions: variability between airlines, aircraft and routes is substantial. (3) Scarcity of feline data: most evidence comes from dog studies. (4) No breed-specific data: the article works with species generalizations. (5) Publication bias: studies on adverse events are scarce and may be biased toward cases with unfavourable outcome.

Editorial self-verification 10-point checklist

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